Agent_4

https://gemini.google.com/app/bc5c27b64e9ff2b9

User prompt: you are part of a big mission and your task is given in the md file

Response: 1. Ground Level Boundary Condition

The vertical potential $V(y)$ contains an infinite hard-wall potential at ground level ($y = 0$):

$$V(y) = \begin{cases} mgy, & \text{for } y \geq 0 \\ \infty, & \text{for } y < 0 \end{cases}$$

To maintain continuity and ensure the wave function vanishes identically in the forbidden region $y < 0$, the vertical component $Y(y)$ of the spatial wave function must satisfy the Dirichlet boundary condition at $y = 0$:

$$Y(0) = 0$$

2. Evaluation of $z(0)$

The 1D Schrödinger equation for $y \geq 0$ is:

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} + mgyY(y) = E_yY(y)$$

By introducing the dimensionless variable $z(y)$:

$$z(y) = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}\left(y - \frac{E_y}{mg}\right)$$

Evaluating $z(y)$ at ground level ($y = 0$):

$$z(0) = \left(\frac{2m^2g}{\hbar^2}\right)^{1/3}\left(0 - \frac{E_y}{mg}\right) = -\left(\frac{2m^2g}{\hbar^2}\right)^{1/3}\frac{E_y}{mg}$$

Moving $E_y/(mg)$ inside the cube root:

$$z(0) = -\left(\frac{2m^2g}{\hbar^2}\cdot\frac{E_y^3}{m^3g^3}\right)^{1/3} = -\left(\frac{2mE_y^3}{\hbar^2g^2}\right)^{1/3}$$

3. Airy Zero Matching

The general solution for the differential equation $\frac{d^2Y}{dz^2} - zY = 0$ is a linear combination of Airy functions:

$$Y(z) = C\mathrm{Ai}(z) + D\mathrm{Bi}(z)$$

Since $\mathrm{Bi}(z) \to \infty$ as $z \to \infty$, physical acceptability (square-integrability as $y \to \infty$) requires $D = 0$, leaving $Y(y) = C\mathrm{Ai}(z(y))$.

Applying the ground boundary condition $Y(0) = 0$ gives:

$$\mathrm{Ai}(z(0)) = 0$$

Let a_n (for $n = 1, 2, 3, \ldots$) represent the n-th negative zero of the Airy function $\mathrm{Ai}(x)$, ordered such that $0 > a_1 > a_2 > a_3 > \ldots$ (where $a_1 \approx -2.3381$, $a_2 \approx -4.0879$, $a_3 \approx -5.5206$).

Matching $z(0)$ to these discrete roots yields:

$$z(0) = a_n$$

4. Exact Quantum Relation for $E_{y,n}$

Equating the expression for $z(0)$ to a_n:

$$-\left(\frac{2mE_{y,n}^3}{\hbar^2 g^2}\right)^{1/3} = a_n$$

Taking the cube of both sides:

$$-\frac{2mE_{y,n}^3}{\hbar^2 g^2} = a_n^3 \implies \frac{2mE_{y,n}^3}{\hbar^2 g^2} = -a_n^3 = |a_n|^3$$

Solving for $E_{y,n}^3$:

$$E_{y,n}^3 = \frac{\hbar^2 g^2}{2m}(-a_n)^3$$

Taking the cube root provides the exact energy eigenvalues for the vertical motion in the gravitational quantum well:

$$E_{y,n} = \left(\frac{\hbar^2 g^2}{2m}\right)^{1/3}(-a_n) = \left(\frac{mg^2\hbar^2}{2}\right)^{1/3}|a_n|$$
